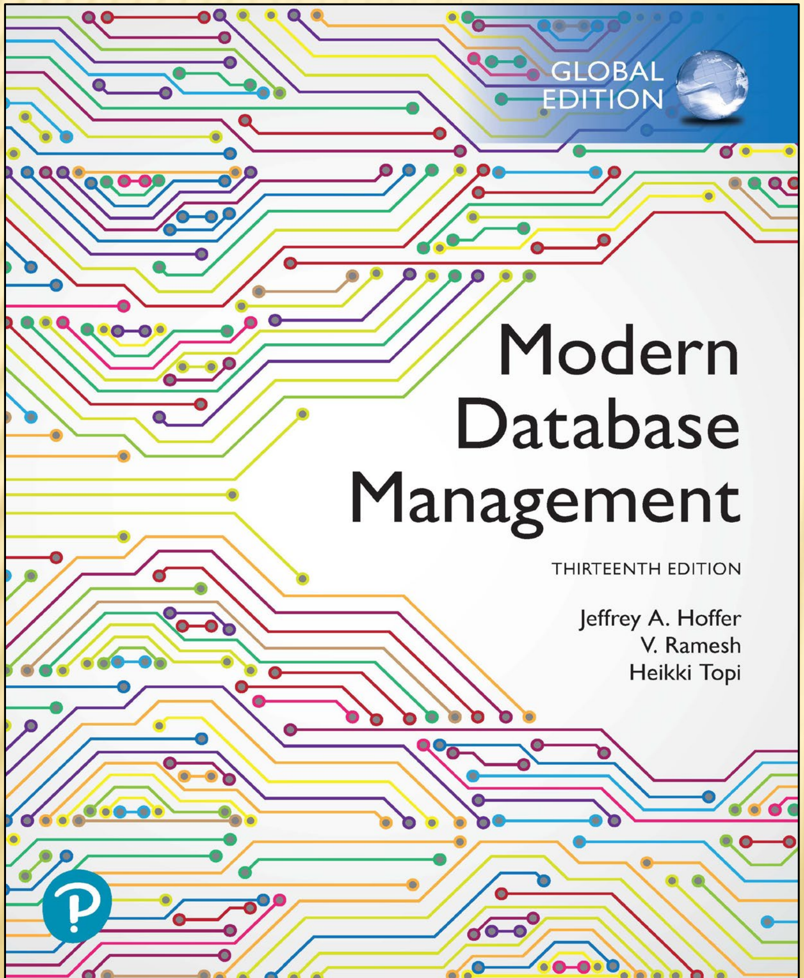
Modern Database Management

Thirteenth Edition



Chapter 7

Databases in Applications

Learning Objectives

7.1 Define terms

7.2 Explain three components of client/server systems: presentation, processing, and storage

7.3 Distinguish between two-tier and three-tier architectures

7.4 Describe key components and information flow in Web applications

7.5 Describe how to connect to databases in 3-tier applications using Java (JSP) and Python

7.6 Understand the notion of transaction integrity

7.7 Compare optimistic and pessimistic concurrency control

7.8 Understand the basics of application security

CLIENT/SERVER ARCHITECTURES

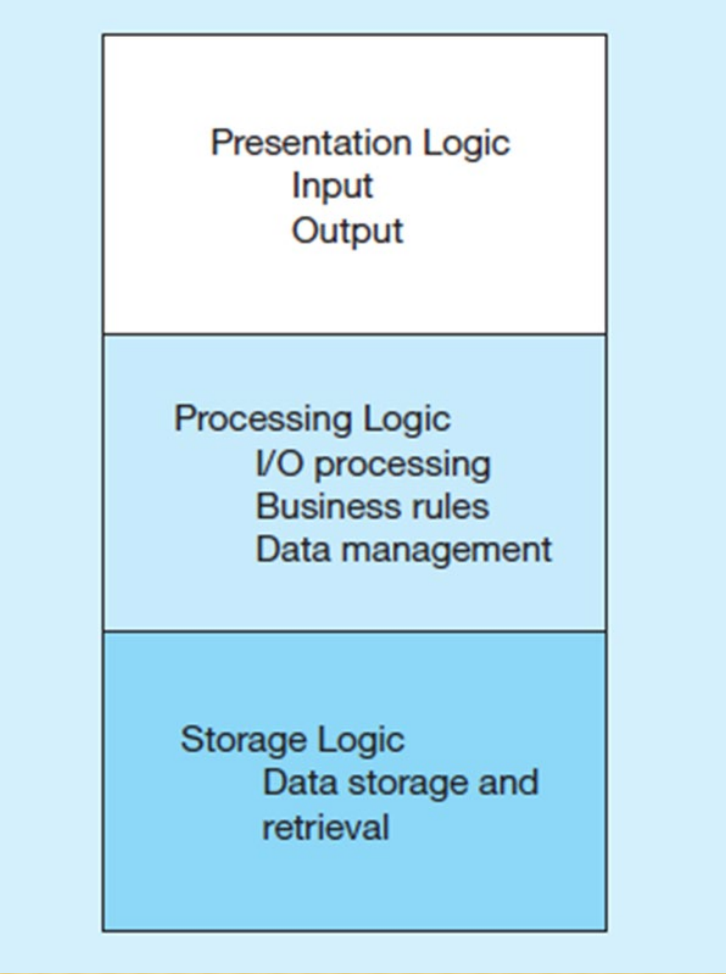
- Networked computing model
- Processes distributed between clients and servers
- Client – Workstation (PC, smartphone, tablet) that requests and uses a service
- Server – Powerful computer (PC/mini/mainframe) that provides a service
- For DBMS, server is a database server
- For the Internet, server is a Web server

Figure 7-1 Application Logic Components

GUI interface (e.g. through a browser)

Procedures, functions, programs

DBMS activities



APPLICATION PARTITIONING

- Placing portions of the application code in different locations (client vs. server) after it is written
- Advantages
  - Improved performance
  - Improved interoperability
  - Balanced workloads

FAT VS. THIN CLIENTS

- Fat client – a client PC that is responsible for processing presentation logic, extensive application and business rules logic, and many DBMS functions
- Thin client – an application where the client (PC) accessing the application primarily provides the user interfaces and some application processing, usually with no or limited local data storage

Figure 7-2 Common Logic Distributions (1 of 2)

a) Two-tier client-server environments

Processing logic could be at client (fat client), server (thin client), or both (distributed environment).

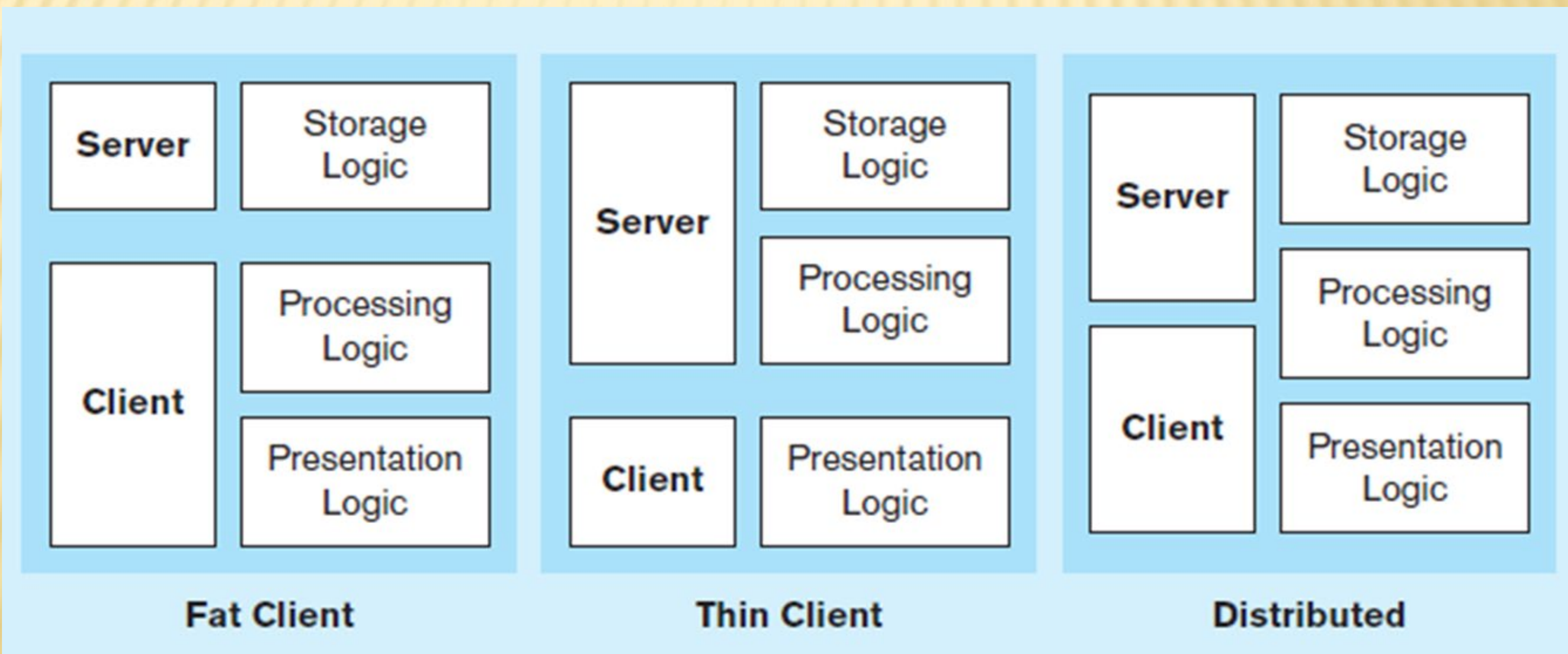
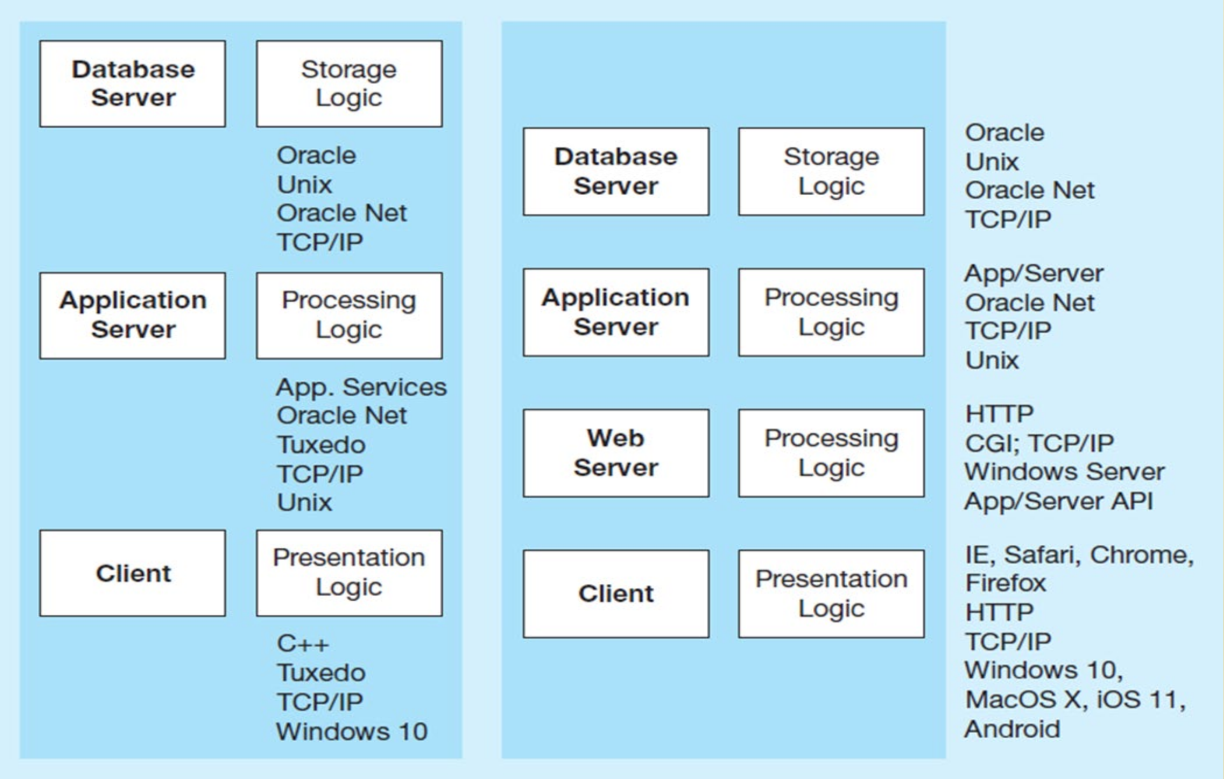


Figure 7-2 Common Logic Distributions (2 of 2)

b) Three-tier and *n*-tier client-server environments

Processing logic will be at application server or Web server.



WEB APPLICATION COMPONENTS

- Database server – hosts the DBMS
  - e.g., Oracle, SQL Server, Informix, MS Access, MySql
- Web server – receives and responds to browser requests using HTTP protocol
  - e.g., Apache, Internet Information Services (IIS)
- Application server – software building blocks for creating dynamic Web sites
  - e.g., MS ASP.NET framework, Java EE, PHP
- Web browser – client program that sends Web requests and receives Web pages
  - e.g., Internet Explorer, Mozilla Firefox, Apple Safari, Google Chrome

Figure 7-3 A Database-Enabled Intranet/Internet Environment

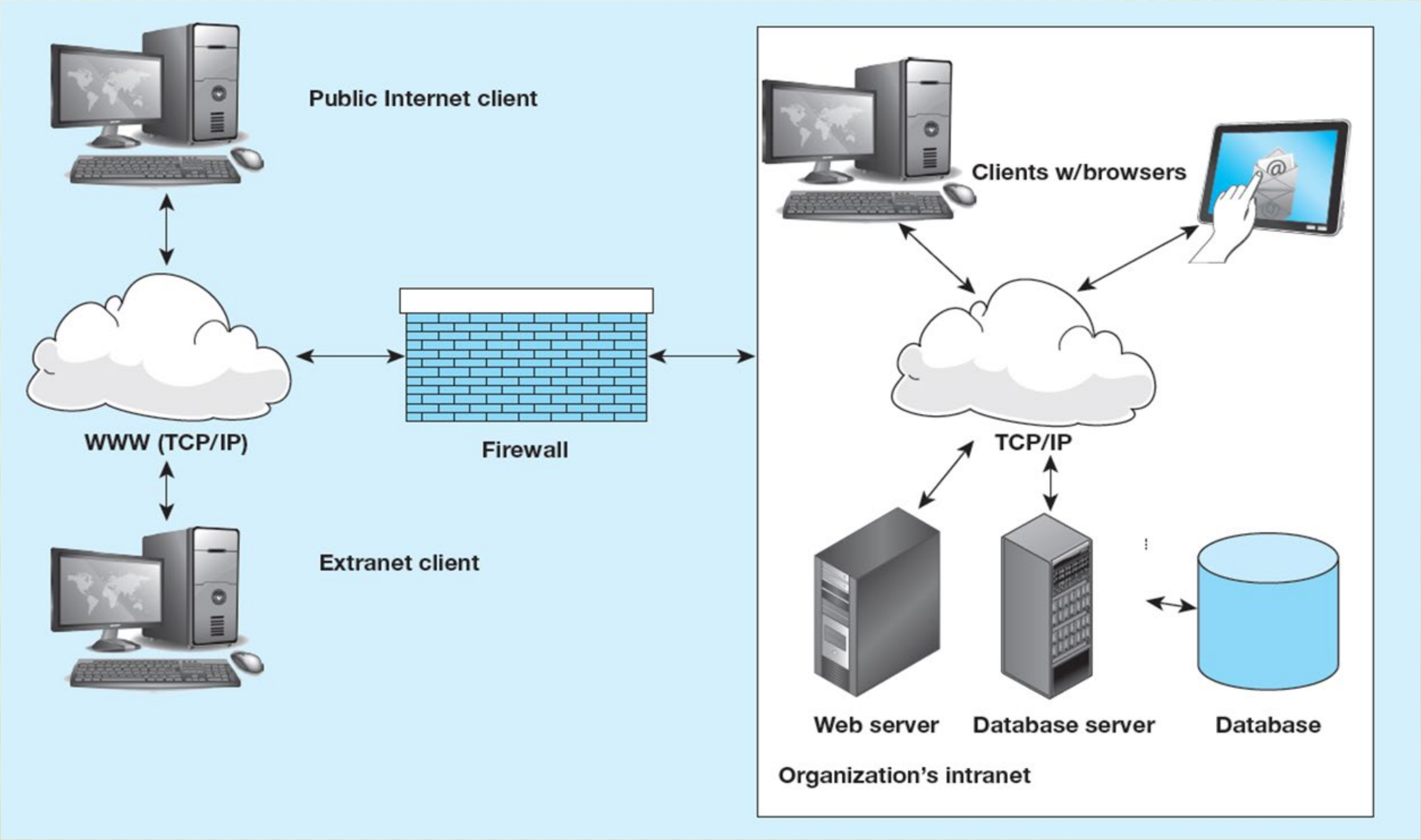
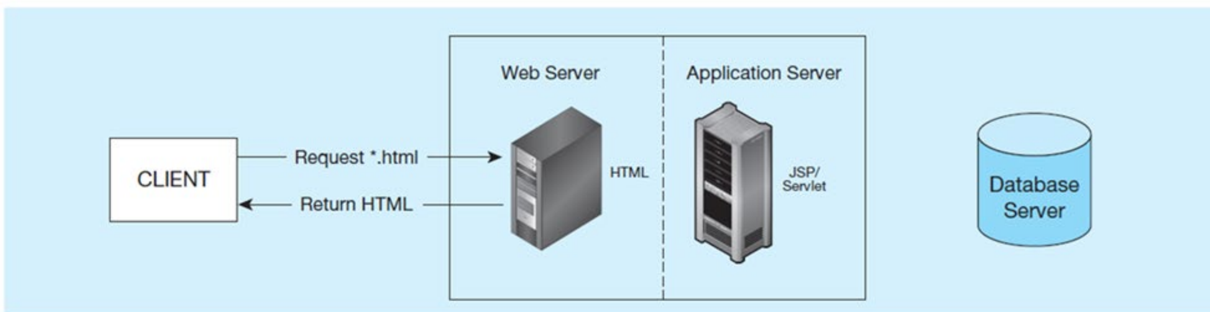


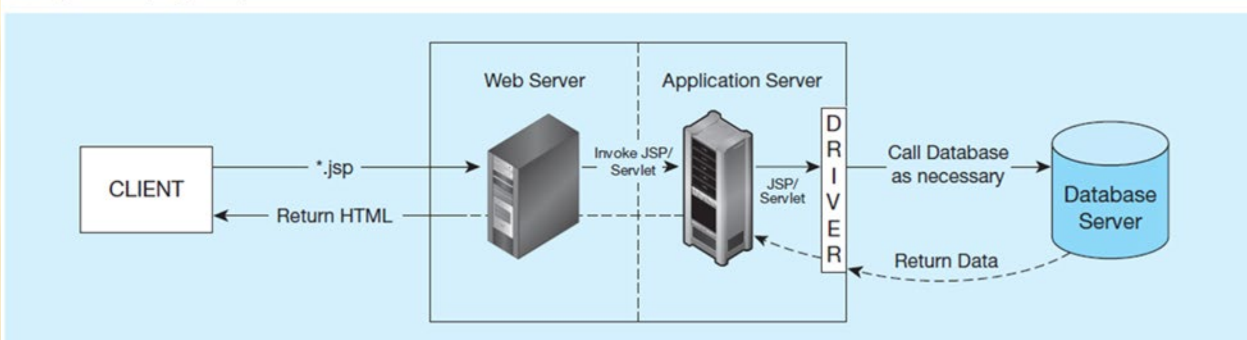
Figure 7-5 Information Flow in a Three-Tier Architecture

(a) Static page request



(b) Has server-side processing, including database access

(b) Dynamic page request



MIDDLEWARE AND APIS

- Middleware – software that allows an application to interoperate with other software without requiring user to understand and code low-level operations
- Application Program Interface (API) – routines that an application uses to direct the performance of procedures by the computer's operating system
- Common database APIs – ODBC, ADO .NET, JDBC

STEPS FOR USING DATABASES VIA MIDDLEWARE APIS

1. Identify and register a database driver.
2. Open a connection to a database.
3. Execute a query against the database.
4. Process the results of the query.
5. Repeat steps 3–4 as necessary.
6. Close the connection to the database.

FIGURE 7-7 DATABASE ACCESS FROM A JAVA PROGRAM

Java implementation of database access to an Oracle database using JDBC

```
import java.sql.*;
public class TestJDBC {
  public static void main(String[ ] args) {
  try {
  Driverd =
  (Driver)Class.forName("oracle.jdbc.driver.OracleDriver").newInstance( );
  System.out.println(d);
  DriverManager.registerDriver (new oracle.jdbc.driver.OracleDriver( ));
  Connection conn =
  DriverManager.getConnection ("jdbc:oracle:thin:@durga.uits.indiana.edu:1
  521:OED1", args[0], args[1]);

  Statement st = conn.createStatement( );
  ResultSet rec = st.executeQuery("SELECT * FROM Student");
  while(rec.next( )) {
    System.out.println(rec.getString("name"));}
  conn.close( );
    }
  catch (Exception e) {
System.out.println("Error – " + e);
  }
  }
}
```

Register the driver to be used.

Identify the type of driver to be used.

Open a connection to a database.

Create a Statement variable that can be used to issue queries against the database

Issue a query and get a result.

Process the result, one row at a time.

Close the connection.

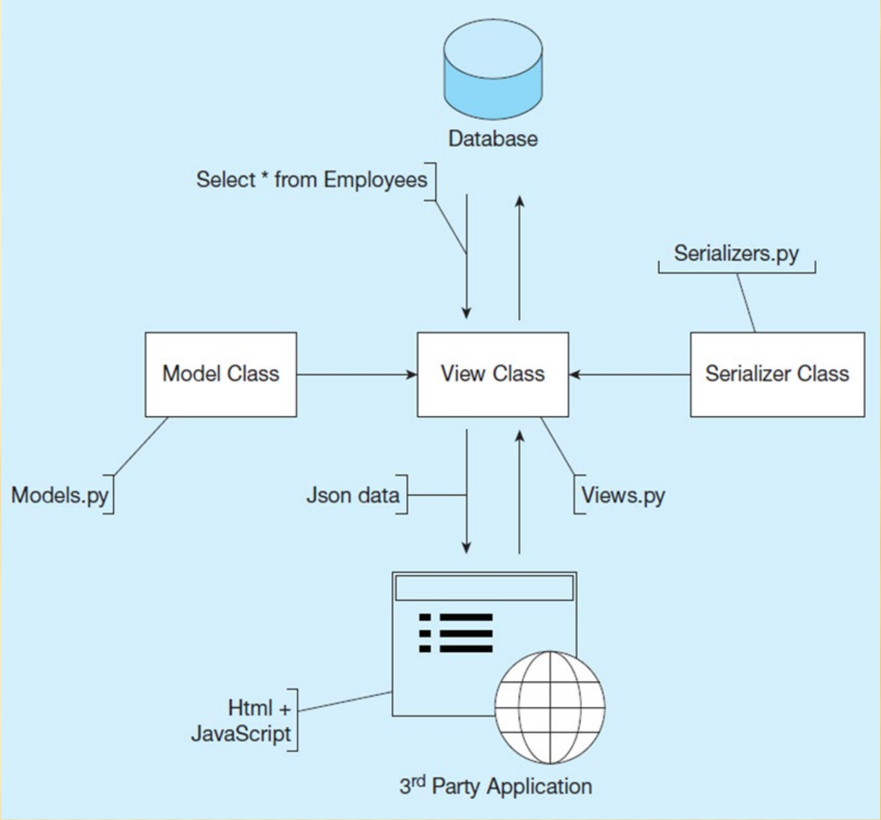
FIGURE 7-8 SYSTEM ARCHITECTURE OF SAMPLE APPLICATION

Python application for database processing.

Model class creates instances of data from table queries.

Serializer class transforms database data into a form that can be used by the client (JSON).

View classes display in browsers for view or edit.



CONSIDERATIONS IN 3-TIER APPLICATIONS

- Stored procedures
  - Code logic embedded in DBMS
  - Improve performance, but proprietary
- Transactions
  - Involve many database updates
  - Either all must succeed, or none should occur
- Database connections
  - Maintaining an open connection is resource-intensive
  - Use of connection pooling

ADVANTAGES AND DISADVANTAGES OF STORED PROCEDURES

- Advantages
  - Performance improves for compiled SQL statements
  - Reduced network traffic
  - Improved security
  - Improved data integrity
  - Thinner clients
- Disadvantages
  - Programming takes more time
  - Proprietary, so algorithms are not portable

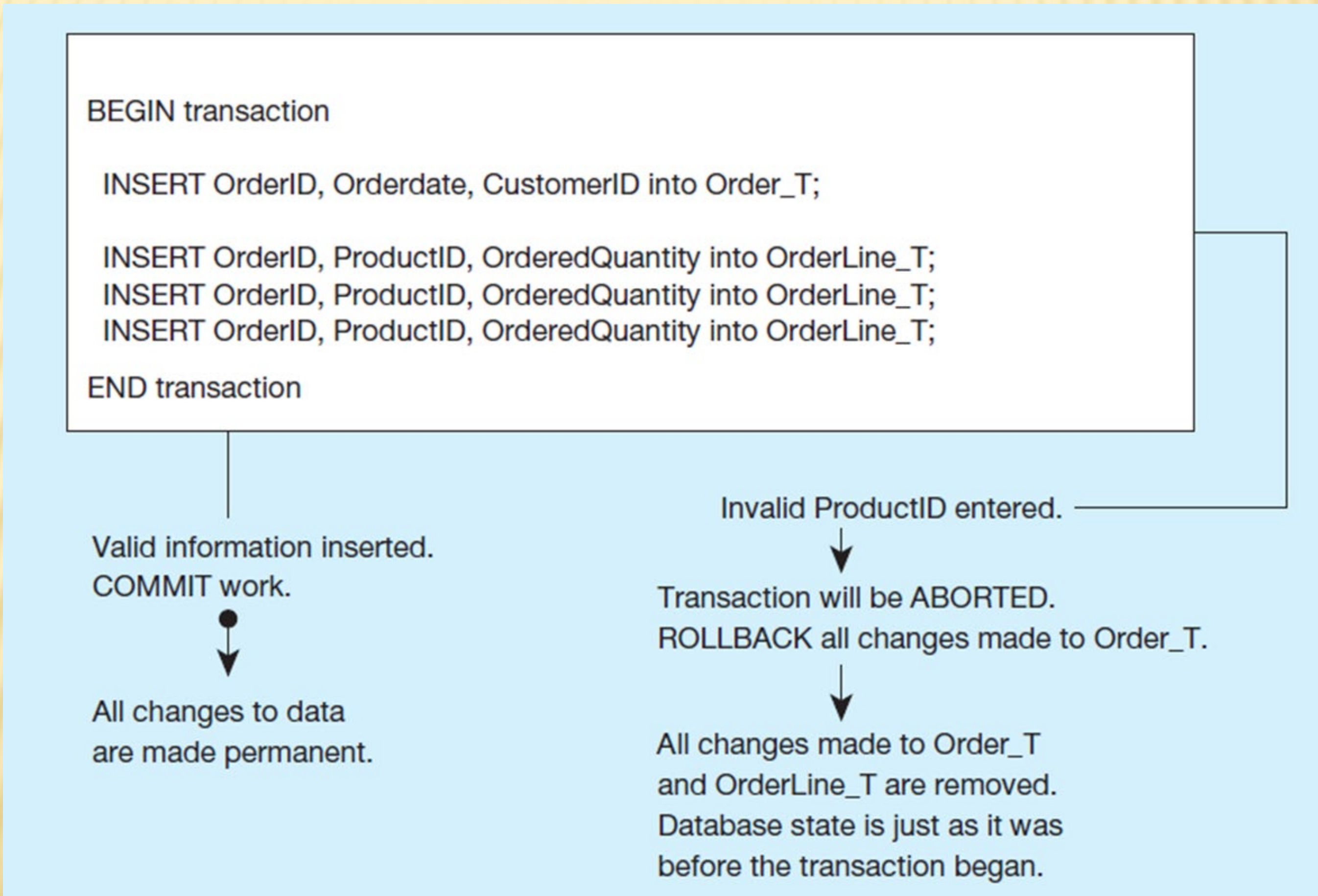
BENEFITS OF THREE-TIER ARCHITECTURES

- Scalability
- Technological flexibility
- Long-term cost reduction
- Better match of systems to business needs
- Improved customer service
- Competitive advantage
- Reduced risk

TRANSACTION INTEGRITY: ACID RULES

- Atomic
  - Transaction cannot be subdivided
- Consistent
  - Constraints don’t change from before transaction to after transaction
- Isolated
  - Database changes not revealed to users until after transaction has completed
- Durable
  - Database changes are permanent

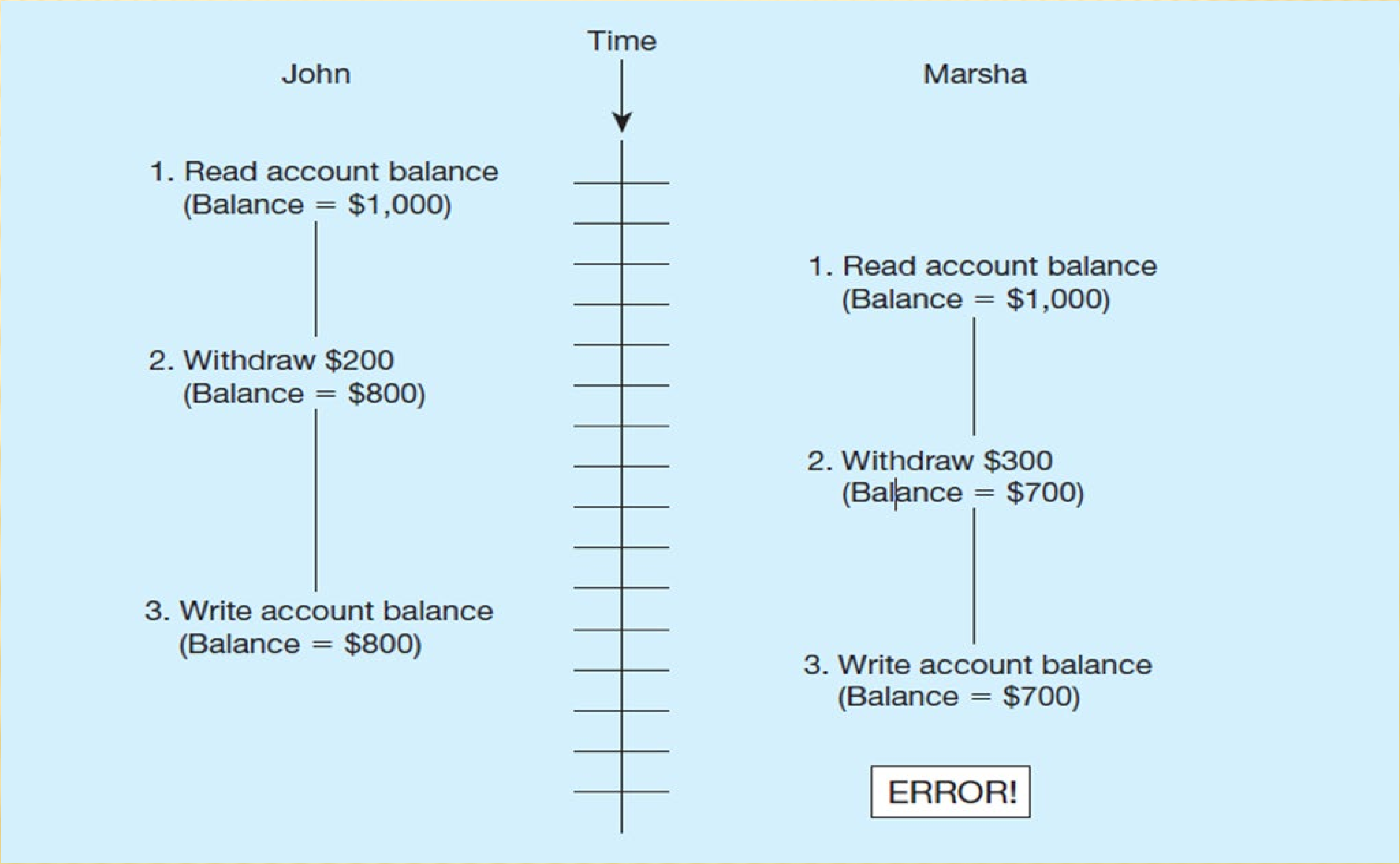
Figure 7-19 A SQL Transaction Sequence (Pseudocode) Environment



CONTROLLING CONCURRENT ACCESS

- Problem
  - In a multi-user environment, simultaneous access to data can result in interference and data loss (lost update problem)
- Solution – Concurrency Control
  - Managing simultaneous operations against a database so that data integrity is maintained and the operations do not interfere with each other in a multi-user environment

FIGURE 7-20 LOST UPDATE (NO CONCURRENCY CONTROL IN EFFECT)

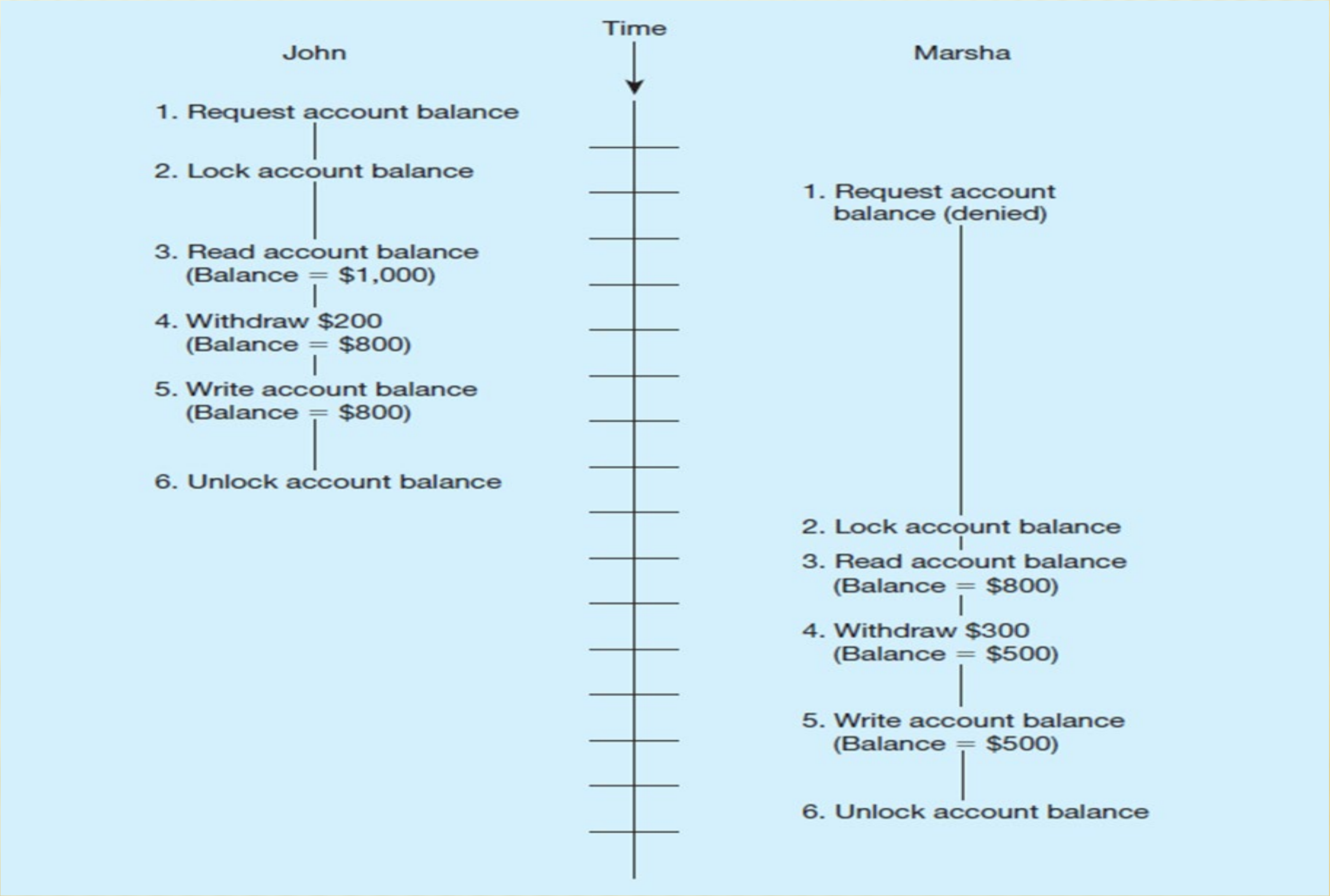


CONCURRENCY CONTROL TECHNIQUES

- Serializability
  - Finish one transaction before starting another
- Locking Mechanisms
  - The most common way of achieving serialization
  - Data that is retrieved for the purpose of updating is locked for the updater
  - No other user can perform update until unlocked

FIGURE 7-21 UPDATES WITH LOCKING (CONCURRENCY CONTROL)

Locking solves the lost update problem

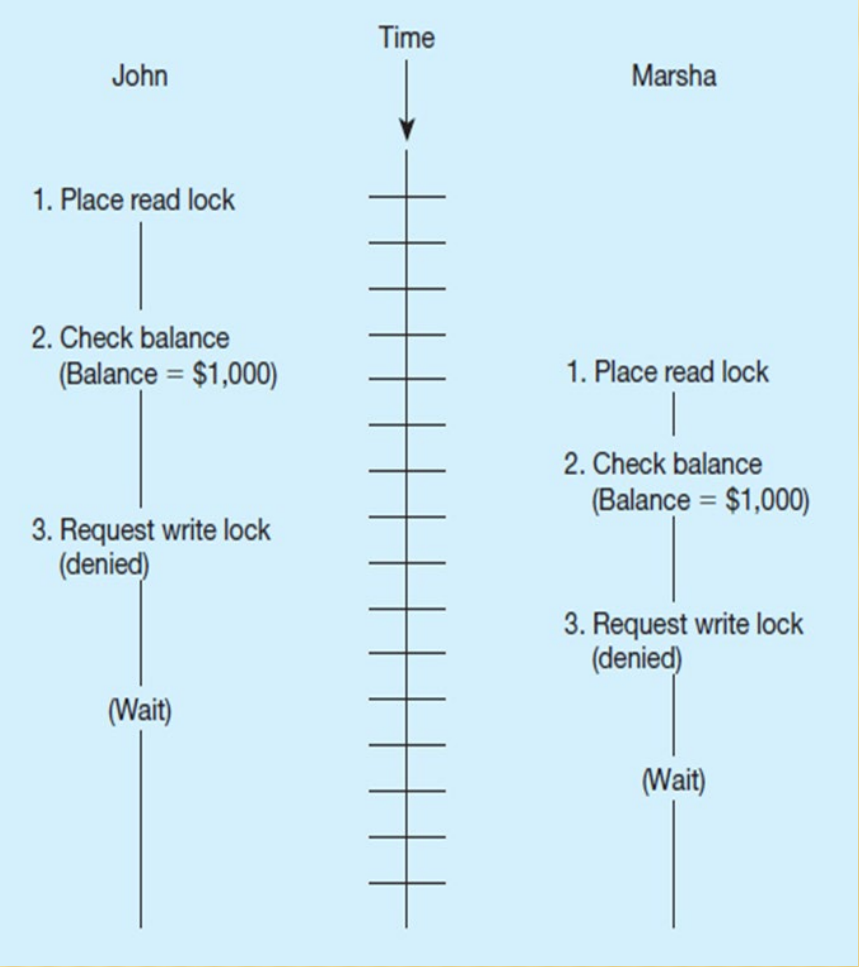


LOCKING MECHANISMS

- Locking level:
  - Database – used during database updates
  - Table – used for bulk updates
  - Block or page – very commonly used
  - Record – only requested row; fairly commonly used
  - Field – requires significant overhead; impractical
- Types of locks:
  - Shared lock – Read but no update permitted. Used when just reading to prevent another user from placing an exclusive lock on the record
  - Exclusive lock – No access permitted. Used when preparing to update

DEADLOCK

- An impasse (僵局) that results when two or more transactions have locked common resources, and each waits for the other to unlock their resources
- Figure 7-22 shows the problem of deadlock
- John and Marsha will wait forever for each other to release their locked resources!



MANAGING DEADLOCK

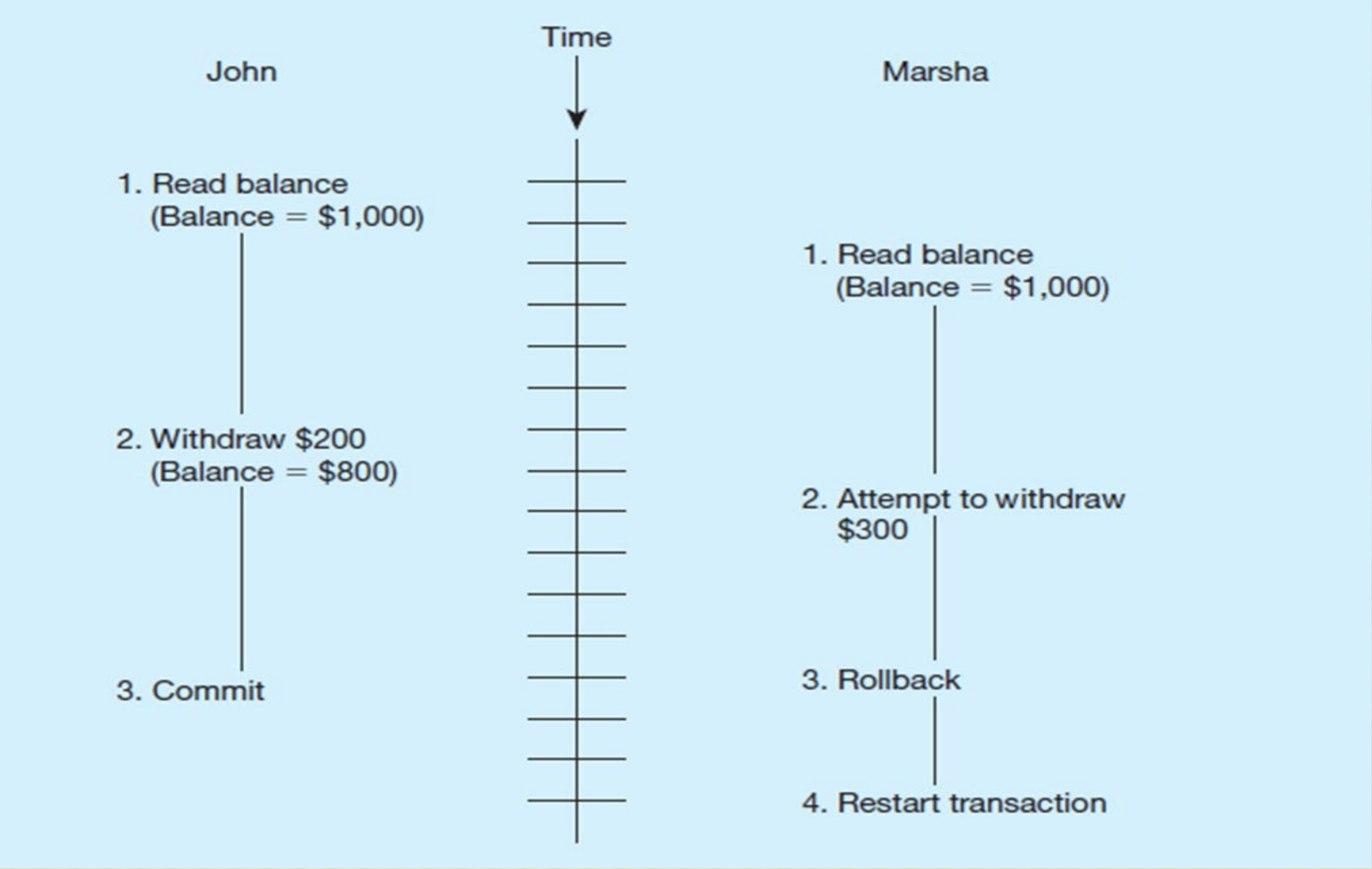
- Deadlock prevention:
  - Lock all records required at the beginning of a transaction
  - Two-phase locking protocol
    - Growing phase
    - Shrinking phase
  - May be difficult to determine all needed resources in advance
- Deadlock Resolution:
  - Allow deadlocks to occur
  - Mechanisms for detecting and breaking deadlock
    - Resource usage matrix
    - Back out (退出) one deadlock at a time
    - Rerun transaction

VERSIONING

- Optimistic approach to concurrency control
- Instead of locking
- Assumption is that simultaneous updates will be infrequent (很少)
- Each transaction can attempt an update as it wishes
- The system will create a new version of a record instead of replacing the old one
- When a conflict occurs, accept one user's update and inform the other user that its update needs to be tried again.
- Use of rollback and commit for this

FIGURE 7-24 THE USE OF VERSIONING

Better performance than locking



DATA SECURITY

- **Database Security:** Protection of the data against accidental or intentional loss, destruction, or misuse
- Increased difficulty due to Internet access and client/server technologies

THREATS TO DATA SECURITY

- Accidental losses attributable to:
  - Human error
  - Software failure
  - Hardware failure
- Theft and fraud
- Loss of privacy or confidentiality
  - Loss of privacy (personal data)
  - Loss of confidentiality (corporate data)
- Loss of data integrity
- Loss of availability (e.g., through sabotage)

FIGURE 7-25 POSSIBLE LOCATIONS OF DATA SECURITY THREATS

Threats come from many sources and vulnerabilities (漏洞) exist in multiple places within an information system.

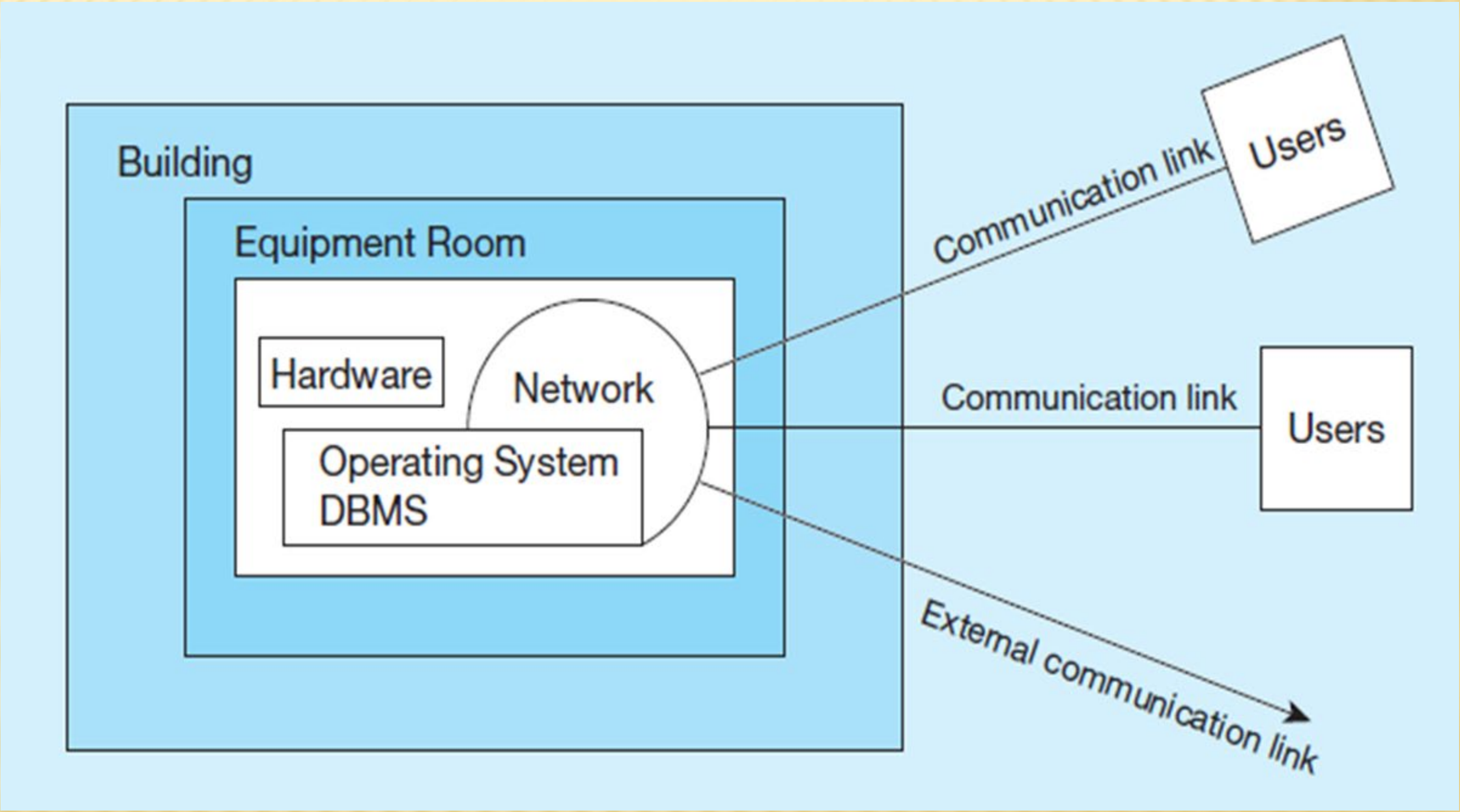
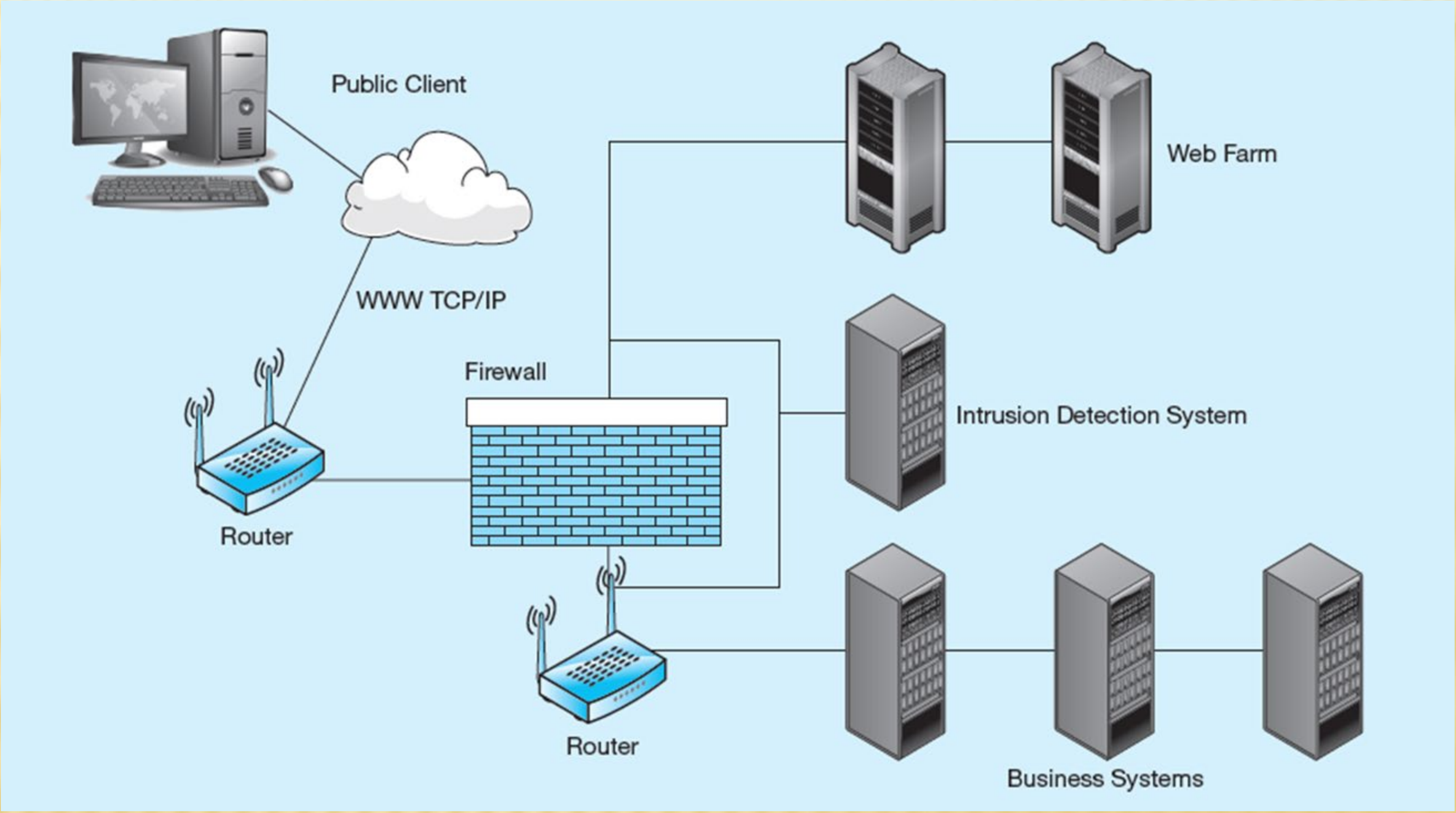


Figure 7-26 Establishing Internet Security



CLIENT–SERVER APPLICATION SECURITY

- Static HTML files are easy to secure
  - Standard database access controls
  - Place Web files in protected directories on server
- Dynamic pages are harder
  - User authentication
  - Session security
  - SSL for encryption
  - Restrict number of users and open ports
  - Remove unnecessary programs

DATA PRIVACY

- W3C Web Privacy Standard
  - Platform for Privacy Protection (P3P)
- Addresses the following:
  - Who collects data
  - What data is collected and for what purpose
  - Who is data shared with
  - Can users control access to their data
  - How are disputes resolved
  - Policies for retaining data
  - Where are policies kept and how can they be accessed